1. a) 930H = 0930H

↓

0000 1001 0011 0000

Descriptor number | TI | RPL

RPL = 00 → Highest privilege level

TI = 0 → Global D. Table

Entry = 0000 1001 0011 0

= 126H

b) Starting address = base address

= D3741000H

As G=1,

Ending address = Start + limit * 4K

= D3791000 + 1524CFFF

= ~~D3756240H~~

E898DFFF

c) Physical Address = Start + offset

= D3791000 + 12ACD

= D3753ACD

d) Access = 11 H

↓

0001 0001

→ Data/Stack

E = 0 ~~→ ES~~

→ ED = 0 → Data Segment

e) A = 1 → So, access granted

f) P = 0 → Descripton undefined

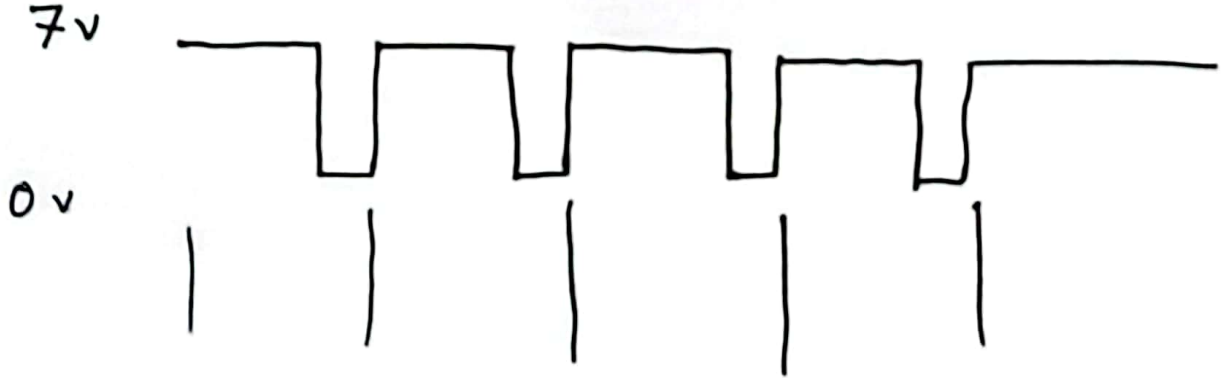
2. a) DAC = 11 bit = 2^{11}

(0 - 2047)

2097 ● → 100 % duty cycle

∴ 1535 ● → $\frac{100 \times 1535}{2097}$ % ..

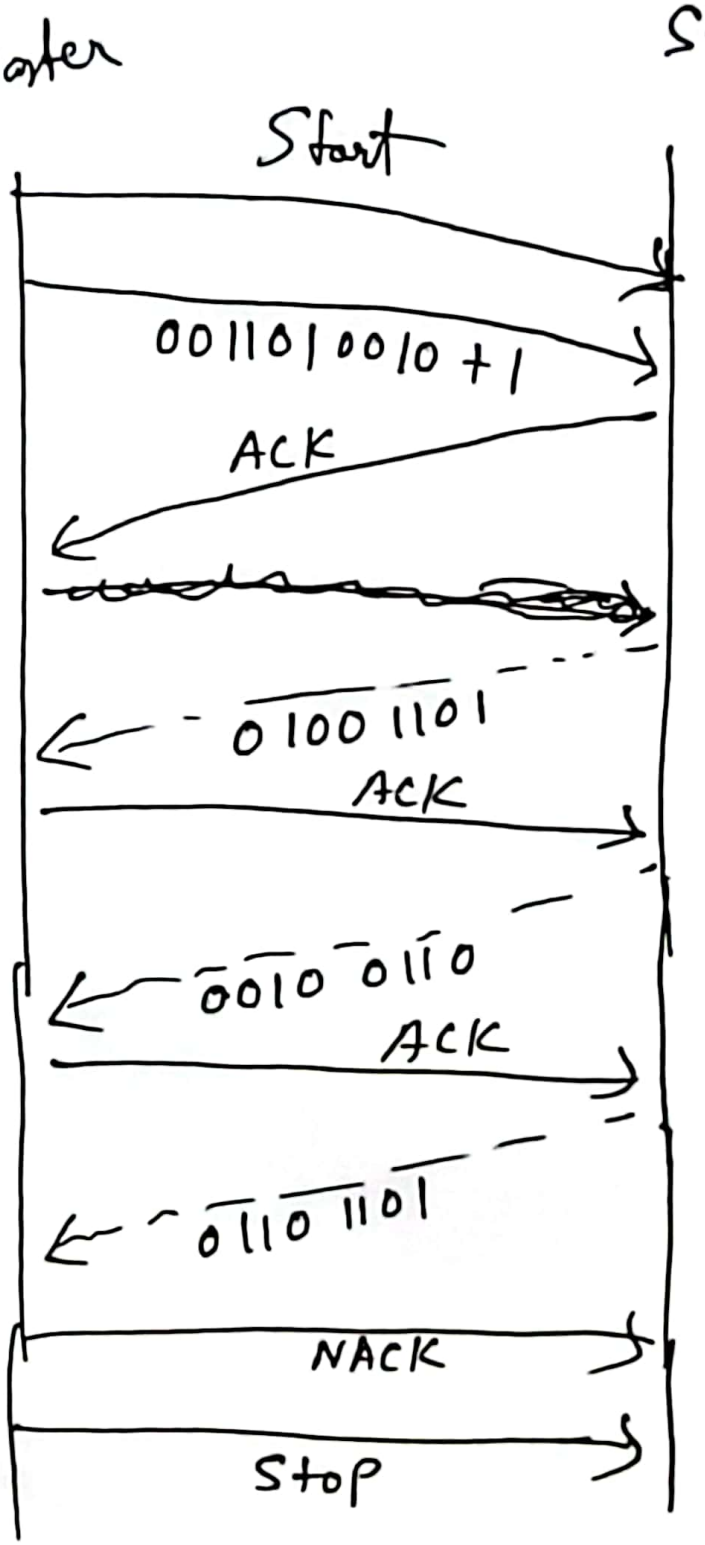
≃ 75 % .



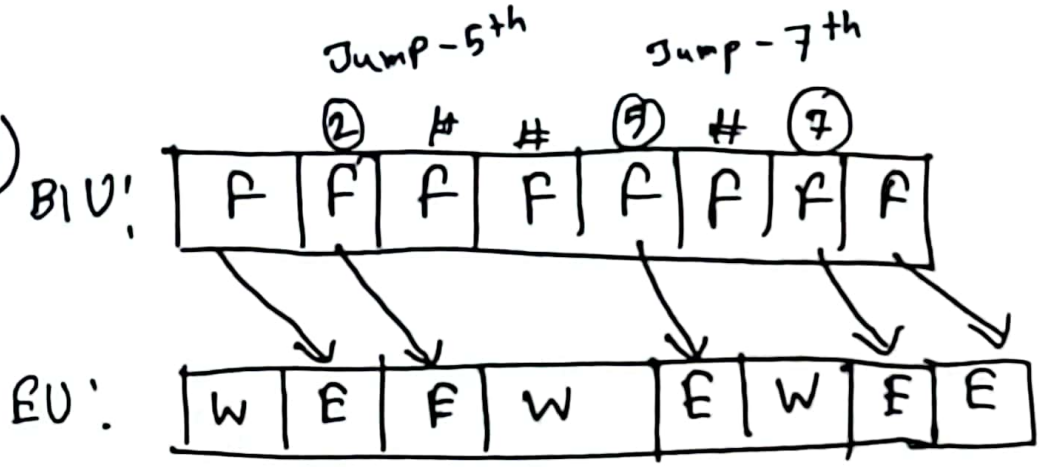
b) No, For entry exit → Arduino

For performance → Raspberry

c)

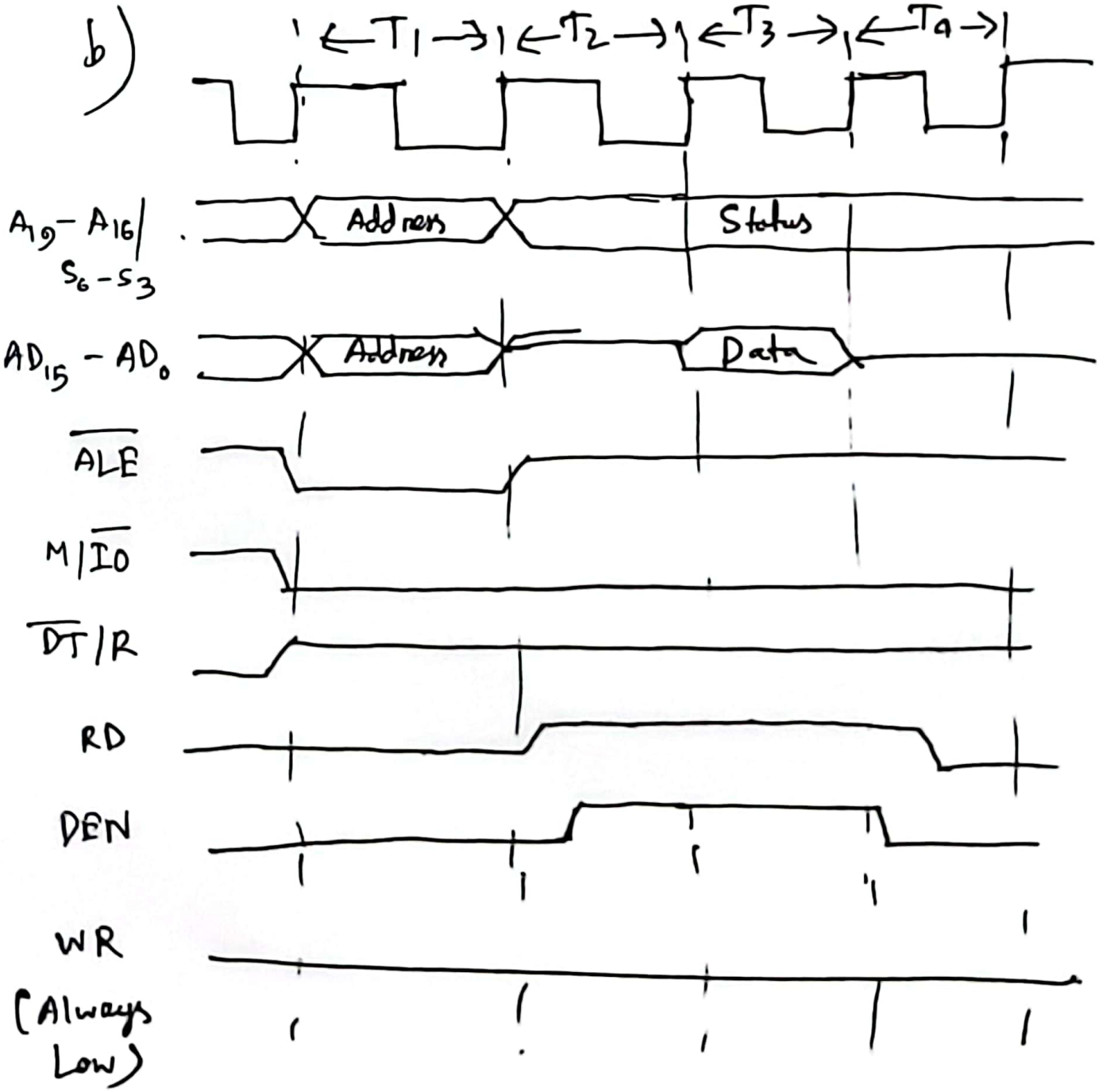


3. a)



discarded ins.

b)



4.a)

```
LDA  FH
ADD  FH
ADD  FH
ADD  FH
SUB  CH
SUB  CH
OUT
HLT
```

AH → 9
BH → 36
CH → 3
DH → 2
EH → 19
FH → 5

b) 0H → 1001 1111
1H → 0110 1111
2H → 0110 1111
3H → 0110 1111
4H → 0100 1100
5H → 0100 1100
6H → 0101 XXXX
7H → 1110 XXXX

c)

Im.	AX
0	5H
1	AH
2	FH
3	14H
4	11H
5	EH
6	EH
7	EH

4. c) Create a new opcode for the instruction OUTB like 1101

T4 (get the RAM address from Instruction register and load it in MAR)

CON = 1110 0110 0100

= E64 H

T5 (get the value from RAM and load it in B register)

CON = 1101 0010 0110

= D26 H

T6 (get the value from adder/subtractor and load it in out register)

CON = 1100 0010 0001

= C21 H